

Wire Gauge Standards

Two of the most commonly used systems for measuring wire thickness are American Wire Gauge (AWG) and Standard Wire Gauge (SWG).

American Wire Gauge (AWG) is a standardized wire sizing system primarily used in the United States and Canada. It dates back to 1857 and provides a numerical value to indicate the diameter of a wire. In AWG, as the number increases, the wire diameter decreases. For instance, a 10 AWG wire is thicker than a 20 AWG wire.

- AWG sizes range from **0000 (4/0)**, the thickest, to **40 AWG**, the thinnest.
- AWG measures the wire's diameter in inches or millimetres.
- Commonly used for electrical wiring, resistance wire, and heating elements.

Standard Wire Gauge (SWG) is a British wire sizing system developed in the 19th century. It's widely used in the UK and other countries that follow metric measurements. Like AWG, SWG assigns numerical values to wire thickness, but the scale differs, as does the wire diameter for each gauge.

- SWG sizes range from **7/0 (largest)** to **50 SWG (smallest)**.
- The diameters are measured in millimetres.
- Commonly used in industrial applications, crafting, and non-electrical uses.

AWG	AWG Diameter (mm)	SWG	SWG Diameter (mm)
10	2.588	10	3.251
12	2.053	12	2.642
14	1.628	14	2.032
16	1.291	16	1.626
18	1.024	18	1.219
20	0.812	20	0.914
22	0.644	22	0.711
24	0.511	24	0.558
26	0.405	26	0.457
28	0.32	28	0.376
30	0.254	30	0.315
32	0.202	32	0.274
34	0.16	34	0.234
36	0.127	36	0.193
38	0.101	38	0.152
40	0.08	40	0.122